

Lab 1 Deliverables

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest cloud billing CSV exports, categorize costs by service and project tag, and highlight compute instances with < 5% CPU utilization over 14 days.	High	Pass: Idle cloud asset flagged with monthly savings estimate; Fail: Active instance misclassified as idle.	(Given) Core functionality to identify wasted compute spend.
FR-002	Functional	The system shall identify and display a list of all unattached or orphaned storage volumes across AWS, Azure, and GCP environments.	High	Pass: Unattached volumes are correctly listed with their respective sizes and costs.	Essential for highlighting unnecessary storage expenses.
FR-003	Functional	The system shall allow the Cloud Administrator to configure and modify monthly budget spending thresholds for different projects.	Medium	Pass: Admin successfully saves a budget limit and the system reflects it.	Required to establish a baseline for budget breach alerts.
FR-004	Functional	The system shall automatically	High	Pass: Alert is sent within 5 minutes of a	Proactive alerting is a core

		dispatch notifications to the FinOps Analyst when current or projected costs breach the configured budget threshold.		budget breach detection.	requirement mentioned in the scenario.
FR-005	Functional	The system shall provide an interactive dashboard displaying aggregated cost data, total potential savings, and a breakdown of costs by cloud provider.	Medium	Pass: Dashboard loads correctly displaying accurate aggregated metrics.	Necessary for the FinOps Analyst to visualize cloud spend across multiple providers.
NFR-001	Nonfunctional	The portal must parse multi-gigabyte cloud billing files and update cost dashboards in under 2 minutes.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	(Given) Performance criteria to ensure smooth user experience with large datasets.
NFR-002	Nonfunctional	The system shall enforce Role-Based Access Control (RBAC) to ensure only authorized Cloud Administrators can modify budget thresholds.	High	Pass: FinOps Analyst is blocked from editing budgets, while Cloud Admin is granted access.	Security constraint to prevent unauthorized modification of financial configurations.

2. Use-Case Flow Specification

Use Case Name: UC-01: Ingest Billing CSV

Primary Actor: FinOps Analyst

Secondary Actor: Cloud Provider System (AWS/Azure/GCP)

System: Multi-Cloud Asset Cost Optimization Portal

1. Preconditions

- The FinOps Analyst is securely logged into the portal.
- The Cloud Provider System has successfully generated the monthly billing CSV export.
- The portal has active storage space to ingest multi-gigabyte files.

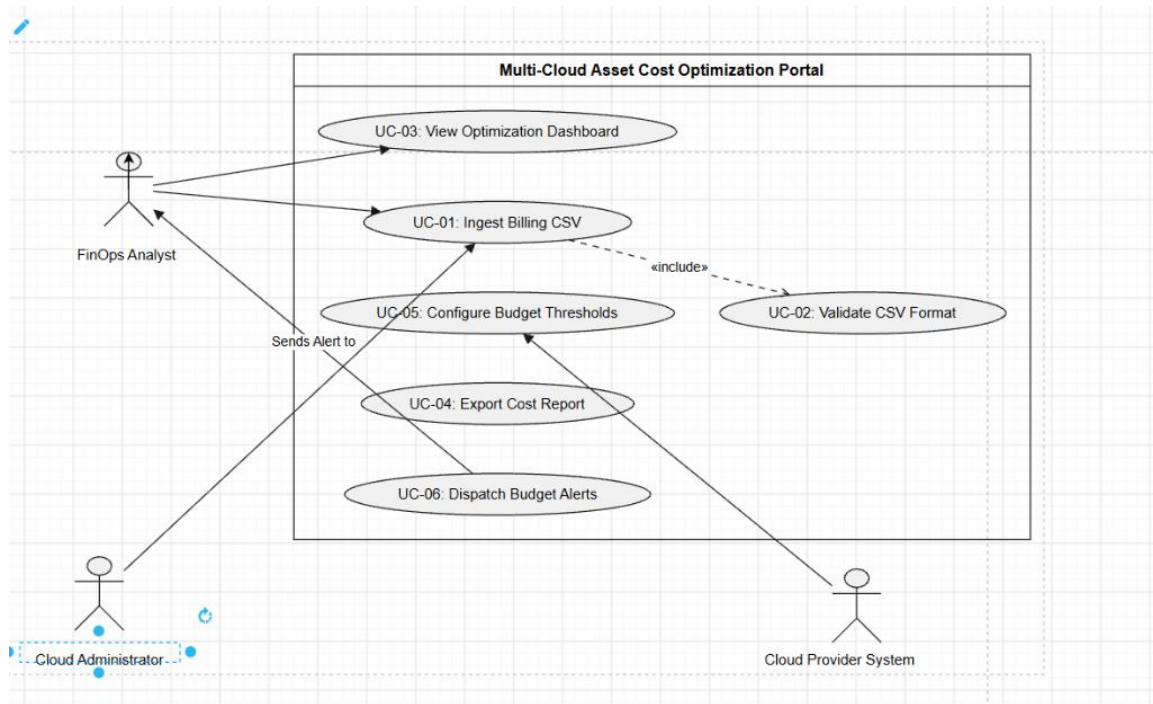
2. Postconditions

- Success: The billing data is successfully ingested, parsed, and categorized. The database is updated, and the Dashboard reflects the latest cost metrics.
- Failure: The system rejects the file, logs an error, and the database remains in its previous state.

3. Main Success Scenario

1. The FinOps Analyst navigates to the 'Data Ingestion' module in the portal.
2. The Analyst clicks 'Upload Billing CSV' and selects the file provided by the Cloud Provider System.
3. The System invokes UC-02: Validate CSV Format to ensure the file headers match expected schemas.
4. The System successfully validates the file format.
5. The System begins parsing the multi-gigabyte file, categorizing costs by service and project tag.
6. The System scans the compute instances and storage data, flagging any compute instances with < 5% CPU utilization over 14 days and identifying unattached storage volumes.
7. The System calculates the potential monthly savings based on the flagged idle resources.
8. The System updates the database and the Cost Optimization Dashboard with the new data.
9. The System displays a success message to the FinOps Analyst.
10. The use case ends successfully.

3. UML Use-Case Diagram



4. Alternate Flow

File Validation Fails (Corrupted or Incorrect Format)

1. During validation (Step 3), the system detects that required columns are missing or corrupted.
2. The System halts the ingestion process.
3. The System displays an error message to the FinOps Analyst.
4. The system logs the validation error for auditing purposes.
5. The use case ends with a failure state, prompting the user to try again.

